

On the Geometry of the Two-Circle Wheatley Leaflet

Bruno Bezerra Trigueiro

Instituto de Ciências Matemáticas e de Computação (ICMC)
Universidade de São Paulo (USP), São Carlos, SP, Brazil

Technical note September 2026

Abstract

We study the leaflet geometry of the two-circle construction of [McKee et al. \(2026\)](#) and its natural extension to N leaflets. Three results may be useful to the authors of that paper.

- (i) *The junction is a crease.* The two halves of each leaflet meet at a fold angle of $41\text{--}47^\circ$ ($49\text{--}53^\circ$ with the proportions of the paper's 15 mm models). For $N \geq 3$ the two cones admit no tangent-continuous junction at all; only $N = 2$ is crease-free.
- (ii) *Comparing different N requires fixing something.* With the total leaflet area fixed, fixing the ring radius or the height gives different shapes; matching the total area of the linear $N = 3$ design while fixing both is impossible for $N = 4$ (certified) and $N = 5, 6, 7$ when $H = R$. In the two equal-area families with the paper's linear profile, the largest fold angle grows with N for $N = 2, \dots, 7$.
- (iii) *Flat sheets.* Each half of a leaflet is a piece of a cone and can be cut from a flat sheet (we give the patterns), but the whole $N = 3$ leaflet (in the paper's normalisation) cannot, not even with a curved fold along the junction. This matters only for manufacture from sheet.

An exploratory finite-element comparison of the sharp and a rounded junction supports no mechanical conclusion. Every result has a written proof or a stated numerical method; selected algebraic identities and geometric results are in addition formalised in Lean 4/mathlib, as listed in Appendix A.

Keywords: developable surfaces; cones; curved folding; heart valve leaflets; polymeric valves; formal verification.

MSC 2020: 53A05, 92C10, 68V20.

Contents

1	Introduction	2
2	The two-circle construction	3
2.1	Conventions and parametrisation	3
2.2	Notes on the source	3
3	Conical structure and developability	4
4	The junction is a crease	6
5	N leaflets	8
6	Area and coaptation	10
7	Comparing different numbers of leaflets	10
7.1	Keeping R and H fixed	11
7.2	The equal-area families	12

8 An exploratory mechanical check: sharp versus rounded junction	12
9 Limitations	13
A Verification table	15

1 Introduction

The reinforced Wheatley valve is a polymeric aortic valve with three leaflets whose shape, in the construction of [McKee et al. \(2026\)](#), is generated by two families of circles: at each height z the section of a leaflet is an arc of a large circle followed by an arc of a small circle. The two principal circles are internally tangent at a commissure; an arc of the large circle and an arc of the small circle rotated through $2\pi/3$ meet at a junction point $M(z)$ to form one leaflet (Figure 1). The construction is explicit, and the authors derive from it a comparison of designs and a finite-element study. In a talk at ICMC-USP (September 2026) the group proposed to compare valves with different numbers of leaflets, $N \in \{2, \dots, 7\}$. This note studies the geometry of the construction with two practical questions in mind: how different values of N can be compared fairly, and whether a leaflet can be cut from a flat polymer sheet.

Contributions.

- Each half of a leaflet is a piece of an oblique circular cone (Theorem 3.1), so it is developable and can be cut flat and bent into place without stretching; we give the flat patterns (Theorem 3.4).
- The whole $N = 3$ leaflet is not developable, not even as a curved fold along the junction, because the geodesic curvatures of the two halves do not match there (Theorem 3.6).
- The junction is a crease with a fold angle of $41\text{--}47^\circ$ (Theorem 4.2); for $N \geq 3$ the two cones have no common tangent plane, so no tangent-continuous junction of them exists (Theorem 4.4).
- The relation $a + b = 1$ of the construction does not depend on N ; the free edges close and, for every N , the projections cover the orifice with multiplicity one almost everywhere (Sections 5 and 6 and Proposition 7.2).
- For comparing N at equal total leaflet area, the area scales as $R^2 A_N^{\text{tot}}(H/R, g)$, where g is the vertical profile; matching the area of the linear $N = 3$ design while keeping R and H fixed is impossible for $N = 4, \dots, 7$ when $H = R$, and the equal-area families at fixed R or fixed H are tabulated (Section 7).

An accompanying repository gives the proofs in Lean, the code and a package with the equal-area families (Data availability).

Relation to earlier work. [McKee et al. \(2021\)](#) built an earlier model of the valve from intersecting and contiguous circles. They noted that it has lines on the leaflets where the derivative jumps, and argued that the valve needs no mould because the surface of each pair of arcs is “topologically equivalent to a laminar sheet”. [Rebolledo et al. \(2023\)](#) removed that kink with a cubic patch, in each horizontal section, between the two intersection points of a large and a small circle. The 2026 construction has the same structure as the 2021 one (a pair of principal circles internally tangent on the unit circle, [McKee et al., 2021](#), eqs. (5)–(6), and a junction where rotated copies meet), with different radii: $a + b = 1$ in 2026, whereas the radii $(1 + b + b^2)/(2 + b)$ and $(1 - b)/2$ of the 2021 circles ([Rebolledo et al., 2023](#), eqs. (3.2)–(3.3)) add up to $1 + b(b - 1)/(2(b + 2)) < 1$ for $0 < b < 1$. The junction curve M of 2026 is therefore the analogue of the 2021 kink, and its existence is not new. [Oliveira et al. \(2022\)](#) generalised the 2021 model to ellipses and showed leaflet designs unfolded onto a plane (their p. 21 and Figs. 11–13), and [Oliveira et al. \(2024\)](#) state that, for manufacturing, the leaflet should be obtainable by cutting a figure from a flat polymer sheet and folding it into position. What is new here, for the 2026 construction, is the size of the crease, the proof that the two cones admit no tangent-continuous junction (with a local rigidity lemma indicating that a developable rounding would have to leave them), and the answer to the flat-sheet question: each half can

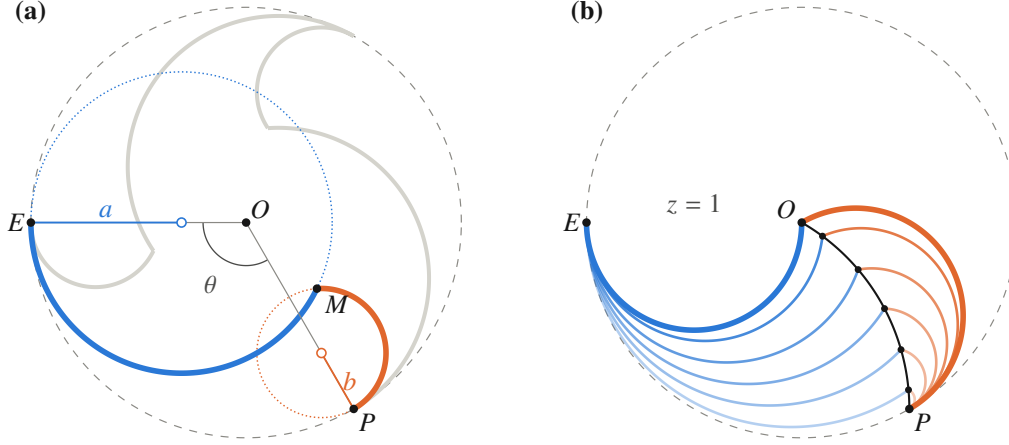


Figure 1: The two-circle construction ($N = 3$). (a) One horizontal section, $z = 0.6$. Surface 1 is the arc EM of the large circle (centre $(-b, 0)$, radius $a = 1 - b$; blue); surface 2 is the arc MP of the small circle (radius $b = z/2$) rotated by $\theta = 2\pi/3$ about O (orange). E and $P = R_\theta E$ are commissures and M is the junction; dotted: the full circles; grey: the sections of the other two leaflets. (b) Sections of one leaflet at $z = 0.1, 0.3, \dots, 0.9$ (light to dark) and the free edge $z = 1$ (thick); the junction $M(z)$ (black) runs from P at $z = 0$ to O at $z = 1$.

be cut flat and bent without stretching, but the whole leaflet cannot. Being topologically a sheet does not imply this.

2 The two-circle construction

2.1 Conventions and parametrisation

We follow the paper's normalisation: ring radius 1, height $z \in [0, 1]$, $b = z/2$, $a = 1 - b$ (McKee et al., 2026, eq. (29) and the proof of Prop. 1). Rotations R_θ about the z axis are counter-clockwise, $E = (-1, 0, 0)$ and O is the origin. For N leaflets let $\theta = 2\pi/N$; the paper is the case $N = 3$. The two pieces of leaflet 0 are

$$X_1(t, z) = (-b + a \cos t, a \sin t, z) \quad (\text{arc } EM; \text{ paper, eqs. (38)–(39)}), \quad (1)$$

$$X_2(t, z) = R_\theta(-a + b \cos t, b \sin t, z) \quad (\text{arc } MP; \text{ eqs. (40)–(41) for } \theta = 2\pi/3), \quad (2)$$

both on the parameter domain $\Omega = \{(t, z) : 0 \leq z \leq 1, \pi \leq t \leq t_M(z)\}$, where $t_M(z)$ is the parameter of the junction point $M(z)$ on *both* circles; for $N = 3$,

$$t_M(z) = 2\pi - \arccos \frac{1 + 2b - 2b^2}{2D}, \quad D = 1 - b + b^2.$$

The arcsine (p. 5, Appendix A plot1) and arccosine (p. 6, plot2) forms of t_M in the paper agree (Theorem 5.1 gives t_M for every N). Leaflet k is $R_{k\theta}(X_1(\Omega) \cup X_2(\Omega))$; $P = R_\theta E$ is the next commissure. Figure 1 shows the construction. Our N -leaflet family keeps the two principal circles tangent at E and rotates the second one by $2\pi/N$; it is our generalisation, not the paper's. Sections 3 and 4 treat $N = 3$ unless stated.

Vertical scale. The mechanical models of the paper are 20 mm in diameter and 15 mm high, whereas $z \in [0, 1]$ with ring radius 1 gives height 10 mm at radius 10 mm. Sections 3–6 use the normalised family (vertical scale $h = 1$) unless stated; Section 7 treats general $h = H/R$. A vertical stretch $(x, y, z) \mapsto (x, y, hz)$ preserves the cone structure and developability, but changes angles, areas and flat patterns (Proposition 4.3).

2.2 Notes on the source

While transcribing the formulas from the rendered pages of McKee et al. (2026) we noticed three points that the authors may wish to know about; none affects the construction.

1. Eq. (28) as printed, $\sin t = -\sqrt{a-b}/(2(a^2+ab+b^2))$, is not consistent with (27) and (29): with it, $\cos^2 t + \sin^2 t - 1 = -(2b-1)(3b-1)/(2D^2)$ on the family $a = 1-b$, which vanishes only at $b = 1/3$ and $b = 1/2$. The intended expression is $-\sqrt{3}(a-b)/(2(a^2+ab+b^2))$, which reappears on pp. 5–6.
2. With counter-clockwise rotations the apex of the second surface is $V_2 = (\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0) = P$, which is $X_2(t, 0)$ for every t . Proposition 1 prints $(\frac{1}{2}, +\frac{\sqrt{3}}{2}, 0)$, obtained by applying (3)–(4) as a point map (a clockwise rotation) to $(-1, 0, 0)$. The set of six apices in (54) is the same in both readings; only the assignment of apices to leaflets changes.
3. In the proof of Proposition 1 the step “ $y = 0$ ” divides by $\sin t_j - \sin t_i$, which vanishes for $t_i = 4\pi/3$, $t_j = 5\pi/3$ (both on EM). The statement remains true (Theorem 3.1).

3 Conical structure and developability

Theorem 3.1 (Proposition 1 of McKee et al., 2026, restated). *For all $(t, z) \in \mathbb{R}^2$,*

$$X_1(t, z) = (1 - \frac{z}{2})(\cos t, \sin t, 0) + \frac{z}{2} V_1, \quad X_2(t, z) = (1 - \frac{z}{2})V_2 + \frac{z}{2} Q(t),$$

with $V_1 = (-1, 0, 2)$, $V_2 = R_\theta E = P$ and $Q(t) = R_\theta(\cos t, \sin t, 2)$, a point of a unit circle at height 2. Hence every stay $t = \text{const}$ of surface i lies on a line through V_i : surface 1 is a piece of the cone over the unit circle with apex V_1 , surface 2 a piece of the cone over the circle of Q with apex V_2 .

Proof. With $b = z/2$ and $a = 1-b$, (1) reads $(-b + (1-b)\cos t, (1-b)\sin t, 2b) = (1-b)(\cos t, \sin t, 0) + b(-1, 0, 2)$. In (2), R_θ is linear and fixes the z axis, so $X_2 = R_\theta((1-b)(-1, 0, 0) + b(\cos t, \sin t, 2)) = (1-b)V_2 + bQ(t)$. For fixed t both expressions are affine in z , and $X_1(t, 2) = V_1$, $X_2(t, 0) = V_2$. $\square$

Both surfaces are therefore *cone maps* $X(t, z) = V + (\alpha + \beta z) w(t)$:

$$\begin{aligned} X_1 &= V_1 + (1 - \frac{z}{2})w_1, & w_1(t) &= (\cos t + 1, \sin t, -2), \\ X_2 &= V_2 + \frac{z}{2}w_2, & w_2(t) &= R_\theta(\cos t + 1, \sin t, 2). \end{aligned} \quad (3)$$

Lemma 3.2 (Cone maps). *Let $X = V + (\alpha + \beta z) w(t)$ with w of class C^2 . Then $X_t = (\alpha + \beta z)w'$, $X_z = \beta w$ and $X_t \times X_z = (\alpha + \beta z)\beta(w' \times w)$. Hence X is an immersion at (t, z) if and only if $\alpha + \beta z \neq 0$, $\beta \neq 0$ and $w'(t) \times w(t) \neq 0$, and at every such point its Gaussian curvature $K = (LN - M^2)/(EG - F^2)$ vanishes.*

Proof. The derivatives are immediate. For K : $X_{zz} = 0$, so $N = 0$; and $X_{tz} = \beta w' = \frac{\beta}{\alpha + \beta z} X_t$ is tangent, so $M = X_{tz} \cdot n = 0$. Thus $LN - M^2 = 0$. $\square$

Lemma 3.3 (The two cones of the leaflet). *For every θ , $c_1 := w'_1 \times w_1 = (-2\cos t, -2\sin t, -(1 + \cos t))$ and $c_2 := w'_2 \times w_2 = R_\theta(2\cos t, 2\sin t, -(1 + \cos t))$, with $|c_1|^2 = |c_2|^2 = 4 + (1 + \cos t)^2 > 0$. Moreover $|w'_1| = 1$ and $|w_1|^2 = 6 + 2\cos t \geq 4$. Consequently X_1 is regular if and only if $z \neq 2$ (so on all of Ω) and X_2 if and only if $z \neq 0$, and both have $K \equiv 0$ at their regular points.*

Proof. Direct computation; for c_2 use that a rotation about the z axis commutes with the cross product. The last sentence follows from Lemma 3.2 with $(\alpha, \beta) = (1, -\frac{1}{2})$ and $(0, \frac{1}{2})$. $\square$

Theorem 3.4 (Flat patterns). *Let $r(t) = \sqrt{6 + 2\cos t}$ and*

$$\varphi(t) = \int_{\pi}^t \frac{\sqrt{4 + (1 + \cos s)^2}}{6 + 2\cos s} ds.$$

The planar maps $D_1(t, z) = (1 - \frac{z}{2})r(t)(\cos \varphi, \sin \varphi)$ and $D_2(t, z) = \frac{z}{2}r(t)(\cos \varphi, \sin \varphi)$ have the same first fundamental form as X_1 and X_2 at every (t, z) ; they are immersions exactly where X_1, X_2 are, hence local isometries there. D_1 is injective on Ω and D_2 on $\Omega \cap \{z > 0\}$, and every sector angle $\varphi(t_M(z))$ lies in $[0, \pi/2]$ (numerically 0.8816, 1.0539, 1.2526 rad at $z = 0, \frac{1}{2}, 1$).

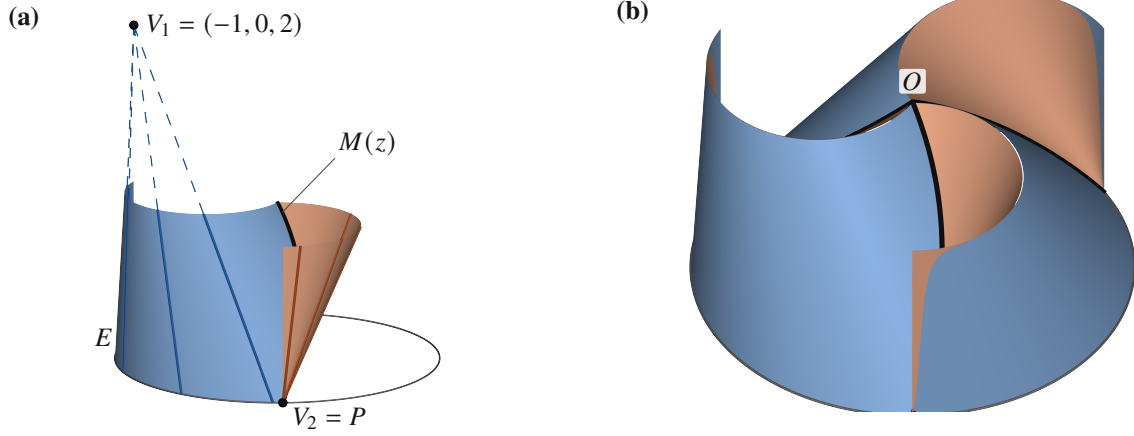


Figure 2: (a) One leaflet ($N = 3$). Surface 1 (blue) is a piece of the cone with apex $V_1 = (-1, 0, 2)$, above the valve: its stays (blue lines) extend (dashed) to V_1 . Surface 2 (orange) is a piece of the cone with apex $V_2 = P$ at the base of the commissure post, where its stays start. Black: the junction $M(z)$. (b) The three-leaflet valve. Rendered from the exact parametrisation with an orthographic camera.

Proof. For a cone map with $r = |w|$ and $\varphi' = |w' \times w|/|w|^2$ put $D = (\alpha + \beta z) r e(\varphi)$, $e = (\cos, \sin)$. Then $D_t = (\alpha + \beta z)(r'e + r\varphi'e^\perp)$ and $D_z = \beta r e$, so $E_D = (\alpha + \beta z)^2(r'^2 + r^2\varphi'^2)$, $F_D = (\alpha + \beta z)\beta r r'$, $G_D = \beta^2 r^2$, while for X , $E = (\alpha + \beta z)^2|w'|^2$, $F = (\alpha + \beta z)\beta w \cdot w'$, $G = \beta^2|w|^2$. Now $rr' = w \cdot w'$ and, by Lagrange's identity, $r^2(r'^2 + r^2\varphi'^2) = (w \cdot w')^2 + |w' \times w|^2 = |w|^2|w'|^2$; so the two forms agree, and so do the determinants $EG - F^2$, which proves the statement about immersions. By Lemma 3.3, $\varphi' = |w' \times w|/|w|^2 \leq |w'|/|w| \leq \frac{1}{2}$ and $\varphi' > 0$; as $t_M(z) - \pi \leq \pi$, the sector angle lies in $[0, \pi/2]$. On that sector φ is strictly increasing, so the polar angle of $D(t, z)$ determines t , and then the radius $(\alpha + \beta z)r(t)$ determines z ($\beta \neq 0$; $1 - \frac{\pi}{2} > 0$, and $\frac{\pi}{2} > 0$ for $z > 0$). $\square$

So each half can be cut from a flat sheet and bent into place without stretching (Figure 3).

Proposition 3.5 (The two cones are congruent). *The isometry $T(x, y, z) = (R_\theta(x, y), 2 - z)$ satisfies $T(X_1(t, z)) = X_2(t, 2 - z)$ for all (t, z) , and $T(V_1) = V_2$. The full stay segments $0 \leq z \leq 1$ with the same t have the same length $\frac{1}{2}\sqrt{6 + 2\cos t}$ on both surfaces. Within Ω this is the stay length for $\pi \leq t \leq 5\pi/3$; for larger t the stay starts at the height $z_{\min}(t)$ where $t_M(z_{\min}) = t$, and its length is $\frac{1}{2}\sqrt{6 + 2\cos t} (1 - z_{\min}(t))$ on both surfaces. The two pieces used in the leaflet are not congruent: surface 1 uses the half of its cone away from the apex, surface 2 the half next to it.*

Proof. Replacing z by $2 - z$ replaces b by $1 - b$, so $X_2(t, 2 - z) = R_\theta(-b + (1 - b)\cos t, (1 - b)\sin t, 2 - z)$, which is $T(X_1(t, z))$; and $T(V_1) = (R_\theta(-1, 0), 0) = V_2$. Along a stay both maps are affine in z with speed $\frac{1}{2}|w_i| = \frac{1}{2}\sqrt{6 + 2\cos t}$ (Lemma 3.3), which gives both lengths. Finally the two pieces have different areas: by Lemmas 3.2 and 3.3, $\text{Area}(X_1(\Omega)) - \text{Area}(X_2(\Omega)) = \frac{1}{2} \int_\Omega (1 - z) \sqrt{4 + (1 + \cos t)^2} dt dz > 0$, so no isometry maps one onto the other. $\square$

Theorem 3.6 (The whole leaflet is not developable). *Let k_1, k_2 be the geodesic curvatures of the junction $M(z)$ as a curve on surface 1 and on surface 2, each measured with the conormal pointing into its own piece. Then*

$$k_1 + k_2 = \frac{4\sqrt{3}(z - 1)\tilde{D}^2}{\sqrt{\tilde{D}^2 + 9}(\tilde{D}^2 + 12)^{3/2}}, \quad \tilde{D} = z^2 - 2z + 4,$$

which is negative for $0 < z < 1$ (value $-5408\sqrt{939}/2146867$ at $z = \frac{1}{2}$) and vanishes at $z = 1$. The concentrated Gaussian curvature carried by the junction is $\int_M (k_1 + k_2) ds \approx -0.10825$ rad. Consequently no two-sided neighbourhood of a point $M(z_0)$, $0 < z_0 < 1$, in the leaflet is intrinsically isometric to a plane region: the leaflet cannot be obtained from a single flat sheet by bending and folding without stretching.

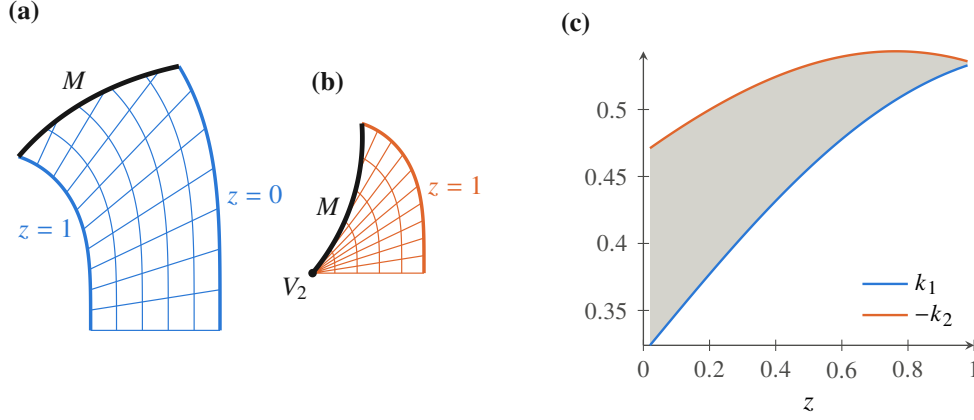


Figure 3: Flat patterns of (a) surface 1 and (b) surface 2 (Theorem 3.4); thin lines are images of $z = \text{const}$ and of stays ($t = \text{const}$), black is the image of the junction M . (c) Geodesic curvature of the junction on each side, measured towards its own piece, computed from the flat patterns. Sewing the two patterns along M would require $k_1 = -k_2$; the shaded gap is $-(k_1 + k_2) > 0$ and agrees with the closed form of Theorem 3.6 to 10^{-4} .

Proof. The formula for $k_1 + k_2$ is a computer-algebra computation from (1)–(2), exact at rational heights; it agrees with the curvatures of the two images of the junction in the flat patterns, computed independently (Figure 3c). An isometry preserves the geodesic curvature of curves (Fuchs and Tabachnikov, 1999, proof of their main theorem). Suppose that a two-sided neighbourhood of $M(z_0)$, where both pieces and the junction are regular, were intrinsically isometric to a plane region. Then an arc of M around $M(z_0)$ would be mapped to a single plane curve with the two pieces on opposite sides of it, so $k_1 = -k_2$ along that arc, contradicting $k_1 + k_2 < 0$ on $(0, 1)$. (At $z = 1$, $k_1 + k_2 = 0$ at a single point, which proves nothing; $z = 0$ is singular.) Equivalently, by Gauss–Bonnet, two flat pieces glued along a curve form an intrinsically flat surface near that curve only if $k_1 + k_2 = 0$. $\square$

Each half, however, can be cut flat (Theorem 3.4). Folding does not help either: a sheet bent and folded without stretching is still intrinsically flat, and $k_1 + k_2 = 0$ along the crease is the condition that the two sides of a curved fold of one sheet must satisfy (Fuchs and Tabachnikov, 1999), so the leaflet is in particular not a curved fold along M . In the flat patterns, the two images of the junction have equal length (1.44932) but different curvature, so they cannot be sewn edge to edge in the plane. This concerns only manufacture from a flat sheet, the route mentioned by Oliveira et al. (2024); moulding or dip-casting are not affected. The Gauss–Bonnet step is classical and not formalised.

4 The junction is a crease

Both pieces lie on the same side $t \leq t_M$ of the junction, so a consistent orientation of the leaflet reverses the normal of one of them. We define the fold angle δ as the angle between $X_{1,t} \times X_{1,z}$ and $-(X_{2,t} \times X_{2,z})$: $\delta = 0$ means tangent-plane continuity.

Lemma 4.1 (Fold angle at a common parameter). *For every θ and $0 < z < 2$, at points of the two surfaces with the same parameter t ,*

$$\cos \delta = \frac{(1 + \cos t)^2 - 4 \cos \theta}{4 + (1 + \cos t)^2}.$$

Proof. By Lemma 3.2, $X_{1,t} \times X_{1,z} = -\frac{1}{2}(1 - \frac{z}{2})c_1$ and $-(X_{2,t} \times X_{2,z}) = -\frac{1}{2} \cdot \frac{z}{2}c_2$; both factors are negative for $0 < z < 2$, so δ is the angle between c_1 and c_2 . By Lemma 3.3, $c_1 \cdot c_2 = -4(\cos t, \sin t) \cdot R_\theta(\cos t, \sin t) + (1 + \cos t)^2 = (1 + \cos t)^2 - 4 \cos \theta$ and $|c_1||c_2| = 4 + (1 + \cos t)^2$. $\square$

Theorem 4.2 (Fold angle, $N = 3$). *For $0 < z \leq 1$,*

$$\cos \delta(z) = \frac{8D^2 + 9}{16D^2 + 9}, \quad D = 1 - \frac{z}{2} + \frac{z^2}{4},$$

so $17/25 \leq \cos \delta \leq 3/4$, i.e. $41.41^\circ \leq \delta \leq 47.16^\circ$, with the minimum at the free edge. The leaflet is C^0 but not G^1 along the whole junction.

Proof. At the junction $1 + \cos t_M = 1 + \frac{1+2b-2b^2}{2D} = \frac{3}{2D}$ and $\cos \theta = -\frac{1}{2}$, so Lemma 4.1 gives $\cos \delta = (\frac{9}{4D^2} + 2)/(\frac{9}{4D^2} + 4) = \frac{8D^2+9}{16D^2+9}$. The function $u \mapsto (8u+9)/(16u+9)$ is decreasing, and $D^2 \in [\frac{9}{16}, 1]$ for $b \in [0, \frac{1}{2}]$, with $D = \frac{3}{4}$ at $z = 1$; this gives the bounds. Since $\cos \delta < 1$ everywhere, the tangent planes never agree along M , while the two pieces do meet along M (Theorem 5.1). $\square$

Proposition 4.3 (Vertical scale). *With vertical scale $h > 0$ (the height multiplied by h), along the junction of the $N = 3$ leaflet, for $0 < z \leq 1$,*

$$\cos \delta_h = \frac{8h^2D^2 + 9}{16h^2D^2 + 9};$$

for $h = 1.5$ it ranges over $[49.17^\circ, 53.13^\circ]$ instead of $[41.41^\circ, 47.16^\circ]$.

Proof. The stretch replaces w_1, w_2 by $(\cos t + 1, \sin t, -2h)$ and $R_\theta(\cos t + 1, \sin t, 2h)$, hence c_1, c_2 by $(-2h \cos t, -2h \sin t, -(1 + \cos t))$ and $R_\theta(2h \cos t, 2h \sin t, -(1 + \cos t))$, and the proof of Lemma 4.1 gives $\cos \delta_h = ((1 + \cos t)^2 - 4h^2 \cos \theta)/(4h^2 + (1 + \cos t)^2)$. Insert $1 + \cos t_M = 3/(2D)$ and $\cos \theta = -\frac{1}{2}$, and evaluate the monotone expression at $D = \frac{3}{4}$ and $D = 1$. $\square$

Can the crease be rounded while keeping the leaflet developable, so that it can still be formed from flat material? The two cones cannot be joined to each other directly (Theorem 4.4 and the remark after it); for a rounding through a strip we only have a local lemma and an expectation, stated below.

Theorem 4.4 (No common tangent plane). *For $N \geq 3$ the two cones have no common tangent plane: there are no stays t_1, t_2 whose tangent planes coincide. For $N = 2$ the stays $t_1 = t_2 = 2\pi$, which form the junction, do share their tangent plane.*

Proof. By Lemma 3.2 the tangent plane of cone i along the stay t is the plane through V_i with normal $c_i(t)$ (the whole stay lies in it). The planes for t_1 and t_2 coincide if and only if $c_1(t_1) \parallel c_2(t_2)$ and $(V_2 - V_1) \cdot c_1(t_1) = 0$. A computation gives $(V_2 - V_1) \cdot c_1(t_1) = 2(1 + \cos(t_1 - \theta))$, so $t_1 \equiv \theta + \pi$ and $c_1(t_1) = (2 \cos \theta, 2 \sin \theta, -(1 - \cos \theta))$. The horizontal parts of $c_1(t_1)$ and $c_2(t_2)$ both have length 2, so parallel means $c_2(t_2) = \pm c_1(t_1)$. With the sign $+$, the horizontal parts give $t_2 \equiv 0$ and the vertical ones $-2 = -(1 - \cos \theta)$, i.e. $\cos \theta = -1$; with the sign $-$, $t_2 \equiv \pi$ and $0 = 1 - \cos \theta$, i.e. $\cos \theta = 1$. Neither holds for $N \geq 3$. For $N = 2$, $\theta = \pi$ and $t_1 = t_2 \equiv 0$ satisfy both conditions. $\square$

A cone has the same tangent plane along a whole stay. At a point where a piece of cone 1 meets a piece of cone 2 with tangent-plane continuity, the common tangent plane would be tangent to cone 1 along a stay and to cone 2 along a stay. Hence, for $N \geq 3$, no G^1 junction between the two cones exists, along any curve.

Lemma 4.5 (Local rigidity; classical). *Let S be a regular surface of class C^∞ with $K \equiv 0$. Suppose S contains an open piece C_0 of a cone, away from its apex, such that the second fundamental form Π of the cone never vanishes on $C_0 \cup \gamma$, and let $\gamma \subset S$ be a curve in the interior of S that bounds C_0 and is transversal to the stays. Then near every interior point of γ , S coincides with that cone.*

The hypothesis $\Pi \neq 0$ cannot be dropped: a non-planar cone may contain planar regions, and a developable surface can leave the cone through one of them.¹ For the two cones of the leaflet it holds at every regular point: $X_{tt} = (\alpha + \beta z)w''$, so the tt coefficient of Π is a non-zero multiple of $w'' \cdot (w' \times w)$, which equals 2 for w_1 and -2 for w_2 ; the other two coefficients vanish (Lemma 3.2). So $\Pi \neq 0$ for $z \neq 2$ on surface 1 and $z \neq 0$ on surface 2.

¹The graph of $y \varphi(x/y - 1)$ with $\varphi(s) = e^{-1/s^2}$ for $s > 0$, $\varphi(s) = 0$ for $s \leq 0$ is a non-planar cone that is planar where $x < y$; on $|x| < 1/4$ it is planar for $1/2 < y < 1$ and is continued for $1 \leq y < 3/2$ by the graph of $\varphi(y - 1)$.

Proof. We use two classical facts for smooth surfaces with $K \equiv 0$ (do Carmo, 1976, §5-8, Propositions 1 and 2, due to Massey): through a parabolic point (where $K = 0$ and $\text{II} \neq 0$) there passes a unique asymptotic line, which is an open segment of a straight line along which the unit normal, hence the tangent plane, is constant; and a maximal such segment never meets a planar point. Since S is smooth and coincides with the cone on C_0 , its second fundamental form at a point $q \in \gamma$ is, by continuity from C_0 , that of the cone at q , which is non-zero by hypothesis (for the leaflet's cones, q lies on a stay that enters C_0 by transversality, away from the apex, where $L \neq 0$); by continuity every point of S near q is parabolic and the asymptotic direction field is continuous there. The asymptotic line of S through a point of C_0 is the stay of the cone through it (uniqueness), so the stays are straight segments of S that cross γ . Let $r \in S$ be close to q on the other side of γ . Its asymptotic segment is straight, with direction close to that of the stay through q , and extends until it leaves a fixed neighbourhood of q in the interior of S (it cannot end at a planar point). Since γ is transversal to the stays, the segment crosses γ at a point q' near q , where it must coincide with the stay through q' . Hence r lies on the extension of a stay, that is, on the cone. $\square$

The smoothness assumption comes from the reference. Versions for lower regularity exist (e.g. Hartman and Nirenberg, 1959), but we have not checked their exact hypotheses.

Remark 4.6 (Expected, not proved). We expect that for $N \geq 3$ no C^∞ developable surface agrees with the leaflet outside a thin strip around the junction M (restricted to heights z in a compact subinterval of $(0, 1)$). By Lemma 4.5, the stays of both cones would continue as straight lines into the strip, and where a continued stay of one cone reaches the other piece the two tangent planes would have to agree, which Theorem 4.4 excludes. Making this precise requires controlling where the continued stays leave the strip; we have not done this.

A G^1 rounding. In each horizontal section $0 < z \leq 1$, replace a neighbourhood of M by the circular arc of radius $\rho(z) = \rho_0 z$ tangent to both section circles. Each section is then G^1 and the family varies smoothly with z , so the surface is G^1 . The rounded strip is not developable: numerically K reaches about -38 (normalised units, $\rho_0 = 0.05$), and for $\rho_0 = 0.1$ it is non-zero at every sample with $\max |K| > 1$. This is consistent with Remark 4.6. The mechanical comparison of Section 8 uses a variant with radius $\sqrt{(\rho_0 z)^2 + (T/R)^2}$ (never thinner than the shell), which is smooth in z , and removes the corner at the base of the commissure post where the rounding cannot be thickened. For $N = 2$ the junction is a common stay with a common tangent plane (Theorem 5.2), and no rounding is needed. Oliveira et al. (2024) report bending of the leaflets near the centre in diastole for the 2021 geometry; whether a crease of this kind plays a role there is not addressed in this note.

5 N leaflets

Theorem 5.1 (The relation $a = 1 - b$ does not depend on N). *Let $\cos \theta \neq 1$. If the large circle $\{(a - 1 + a \cos t, a \sin t)\}$ and the small circle $\{(b - 1 + b \cos t, b \sin t)\}$ rotated by θ share a point with the same parameter t on both, then $a + b = 1$. Assume further $0 < \theta \leq \pi$ (true for $\theta = 2\pi/N$), $0 \leq b \leq \frac{1}{2}$ and $a = 1 - b$. Then the two surfaces meet at the same parameter*

$$t_M^\theta = 2\pi - \arccos \frac{2ab - (a^2 + b^2) \cos \theta}{\Delta}, \quad \Delta = a^2 + b^2 - 2ab \cos \theta > 0,$$

which equals t_M for $N = 3$. The junction runs from $t_M^\theta = \pi + \theta$, $M = R_\theta E$, at $z = 0$ to $t_M^\theta = 2\pi$, $M = O$, at $z = 1$.

Proof. Write $e = (1, 0)$, $u = (\cos t, \sin t)$ and $v = e + u$. The two points are $p = av - e$ and $q = bv - e$, and $p = R_\theta q$, so $|p| = |q|$. Since $|sv - e|^2 = 1 - s(1 - s)|v|^2$, this gives $(a - b)(1 - a - b)|v|^2 = 0$. If $v = 0$ then $p = q = -e$ and $-e = R_\theta(-e)$ forces $\cos \theta = 1$, which is excluded. If $a = b$ then $p = q = R_\theta p$, so $p = 0$, i.e. $av = e$; hence $\sin t = 0$, $\cos t = 1$, $v = 2e$ and $a = b = \frac{1}{2}$. In every remaining case $a + b = 1$.

For the junction, note first that $\Delta = (a - b)^2 + 2ab(1 - \cos \theta) > 0$ for $b \in [0, \frac{1}{2}]$, $a = 1 - b$ and $\cos \theta < 1$ (if $b = 0$ then $\Delta = 1$). Let $c = (2ab - (a^2 + b^2) \cos \theta)/\Delta$ and $s = (b^2 - a^2) \sin \theta/\Delta$. With $a = 1 - b$

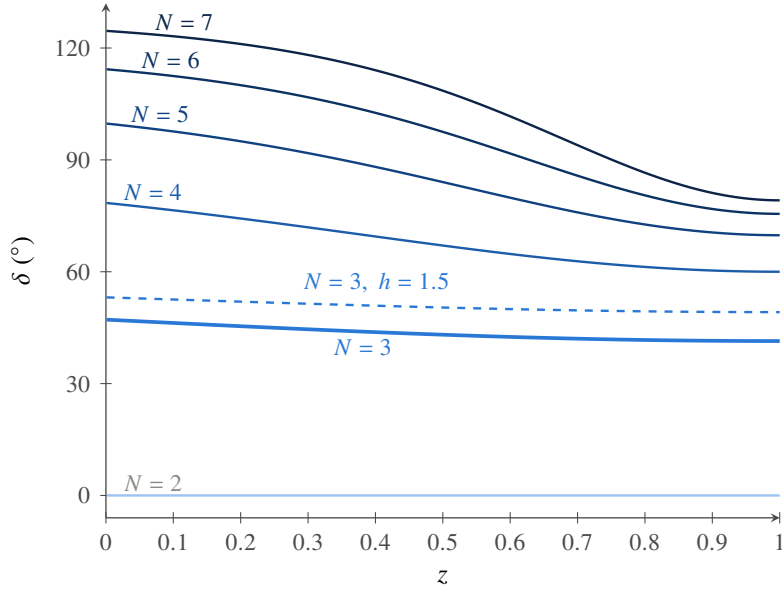


Figure 4: Fold angle δ along the junction for $N = 2, \dots, 7$ (vertical scale $h = 1$, Theorem 5.2) and for $N = 3$ with $h = 1.5$ (dashed, Proposition 4.3). For $N \geq 5$ the fold exceeds 90° near the base; for $N = 2$ there is no crease.

one checks $c^2 + s^2 = 1$ and that (1) and (2) coincide at $(\cos t, \sin t) = (c, s)$. For $0 < \theta \leq \pi$ and $b \leq \frac{1}{2}$, $s \leq 0$, so $t = 2\pi - \arccos c \in [\pi, 2\pi]$. At $z = 0$ ($a = 1, b = 0$), $(c, s) = (-\cos \theta, -\sin \theta)$, so $t = \pi + \theta$ and $M = (\cos(\pi + \theta), \sin(\pi + \theta)) = R_\theta E$; at $z = 1$ ($a = b = \frac{1}{2}$), $(c, s) = (1, 0)$, so $t = 2\pi$ and $M = O$. For $\theta = 2\pi/3$, $c = (1 + 2b - 2b^2)/(2D)$, the paper's value. $\square$

Theorem 5.2 (Fold angle for N leaflets). *For $0 < z \leq 1$, along the junction, with $X = 1 + \cos t_M^\theta = (1 - \cos \theta)/\Delta$,*

$$\cos \delta = \frac{X^2 - 4 \cos \theta}{X^2 + 4}.$$

Hence the junction is a crease if and only if $\cos \theta \neq -1$, i.e. for every $N \geq 3$. For $N = 2$ the junction is the common stay $t = 2\pi$ and the leaflet is G^1 . At the free edge $\cos \delta = \sin^2(\pi/N)$.

Proof. With $a + b = 1$, $1 + c = ((a^2 + b^2)(1 - \cos \theta) + 2ab(1 - \cos \theta))/\Delta = (1 - \cos \theta)/\Delta$; insert this in Lemma 4.1. Then $\cos \delta < 1$ if and only if $\cos \theta > -1$. For $\theta = \pi$, $c = 1$ for every z , so $t_M \equiv 2\pi$ and $\cos \delta = 1$. At $z = 1$, $\Delta = \frac{1}{2}(1 - \cos \theta)$ and $X = 2$, so $\cos \delta = (1 - \cos \theta)/2 = \sin^2(\theta/2)$. $\square$

Theorem 5.3 (The free edges close for every N). *At $z = 1$, $X_2(t, 1) = R_\theta X_1(t, 1)$ for all t : the second piece of leaflet k coincides pointwise with the first piece of leaflet $k + 1$. The free edges are N semicircles of radius $\frac{1}{2}$ joining the centre to the commissures (Figure 5).*

Proof. At $z = 1$, $a = b = \frac{1}{2}$, so both (1) and the argument of R_θ in (2) equal $(-\frac{1}{2} + \frac{1}{2} \cos t, \frac{1}{2} \sin t, 1)$, the circle of radius $\frac{1}{2}$ centred at $(-\frac{1}{2}, 0, 1)$; $t \in [\pi, 2\pi]$ runs over the half from E to O . $\square$

The cone congruence $T(X_1(t, z)) = X_2(t, 2 - z)$ of Proposition 3.5 holds for every θ with the same proof. At $z = 1$ all free edges meet at the centre, so for $N \geq 4$ non-neighbouring leaflets touch there. Below the free edge, at the 25 sampled heights $z \in [0.02, 0.98]$, non-neighbouring leaflets do not touch ($N = 2, \dots, 7$).

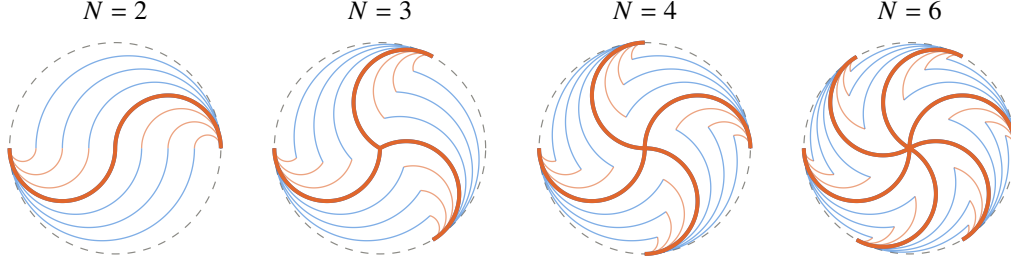


Figure 5: Top views of the N -leaflet valves: sections at $z = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ (thin) and free edges $z = 1$ (thick), which close for every N (Theorem 5.3). Surface 1 blue, surface 2 orange.

6 Area and coaptation

Proposition 6.1 (Area). *For $0 \leq z \leq 2$, $|X_{1,t} \times X_{1,z}| + |X_{2,t} \times X_{2,z}| = \frac{1}{2}\sqrt{4 + (1 + \cos t)^2}$, independently of z and N . Hence the area of one leaflet is*

$$A_N = \frac{1}{2} \int_0^1 \int_{\pi}^{t_M^{\theta}(z)} \sqrt{4 + (1 + \cos t)^2} dt dz,$$

with $A_3 \approx 2.89364$ (radius 1, $h = 1$). Numerically, the total area NA_N increases with N for every tested value $N = 2, \dots, 30$: 7.305, 8.681, 9.889, 10.938, 11.856, 12.668 for $N = 2, \dots, 7$, and 21.237 for $N = 30$. We have no proof for every N .

Proof. By Lemmas 3.2 and 3.3, $|X_{1,t} \times X_{1,z}| = \frac{1}{2}(1 - \frac{z}{2})|c_1|$ and $|X_{2,t} \times X_{2,z}| = \frac{1}{2} \cdot \frac{z}{2}|c_2|$, and $|c_1| = |c_2| = \sqrt{4 + (1 + \cos t)^2}$; the coefficients add up to $\frac{1}{2}$ for $0 \leq z \leq 2$. Both pieces are parametrised over the same Ω . The values are quadratures. $\square$

So in this natural N -family the total leaflet area is *not* preserved. The family $G_N(\alpha)$ of the talk is not defined precisely enough for us to test the claim made there.

Proposition 6.2 (Coaptation). *Let $0 \leq z < 1$. For each pair of facing pieces, piece 2 of leaflet k and piece 1 of leaflet $k + 1$, the sections lie on two circles internally tangent at their common commissure, so they meet only there; at $z = 1$ they coincide along the whole free edge. (For $N = 2$ the two leaflets face each other across both commissures.) The geometric orifice area of the unloaded design is zero.*

Proof. Piece 2 of leaflet k is R_{θ} of the circle with centre $(-a, 0)$ and radius b , and piece 1 of leaflet $k + 1$ is R_{θ} of the circle with centre $(-b, 0)$ and radius a . Both circles pass through E and their centres are $a - b$ apart, the difference of the radii; for $a \neq b$ ($z < 1$) they are internally tangent at E and have no other common point. At $z = 1$ use Theorem 5.3. The projections cover the disk without gaps (Proposition 7.2), so the unloaded orifice area is zero. $\square$

The other pairs of pieces were only sampled. For $N = 2, \dots, 7$ we took 4000 parameter values per section at 25 equally spaced heights $z \in [0.02, 0.98]$, discarding points within distance 0.02 of any commissure post ($R = 1$); the minimum distance between retained sample points of non-facing pieces of neighbouring leaflets was about 0.0121. Together with the sampled check of non-neighbouring leaflets (Section 5), these observations are consistent with zero free coaptation area and height, excluding the attachment posts. They do not establish the absence of contacts between the samples or in the excluded regions.

7 Comparing different numbers of leaflets

A study that compares $N = 2, 3, 4, \dots$ needs a rule saying what is kept fixed. We keep the *total leaflet area* fixed and treat the opening area as a response. Here R is the ring radius and H the height; the section at height Hs uses the parameter $z = g(s)$, where the vertical profile g belongs to $\mathcal{G} = \{g \in C^1([0, 1]) : g' \geq 0, g(0) = 0, g(1) = 1\}$ (the paper uses $g(s) = s$).

Proposition 7.1 (Area density and scaling). *For $0 \leq g \leq 2$, $|X_{1,t} \times X_{1,s}| + |X_{2,t} \times X_{2,s}| = \frac{R}{2} \sqrt{4H^2 + R^2 g'(s)^2 (1 + \cos t)^2}$; for $0 \leq g \leq 1$ the two pieces form the leaflet, and we use only that range (for $N \geq 3$ and larger g the pieces no longer meet along the junction; for $N = 2$ they still do). Hence the total area of the N leaflets is $R^2 A_N^{\text{tot}}(H/R, g)$, with $A_N^{\text{tot}} = N A_N$ in the notation of Proposition 6.1: every dimensionless quantity (fold angle, opening area/ πR^2) depends only on N , g and $h = H/R$. Fixing the area together with R , or together with H , selects different shapes of this family, not the same shape at two scales.*

Proof. A direct computation gives $|X_{1,t} \times X_{1,s}| = \frac{R}{2} (1 - \frac{g}{2}) \sqrt{4H^2 + R^2 g'^2 (1 + \cos t)^2}$ and $|X_{2,t} \times X_{2,s}| = \frac{R}{2} \cdot \frac{g}{2} \sqrt{4H^2 + R^2 g'^2 (1 + \cos t)^2}$. Integrating over $\{0 \leq s \leq 1, \pi \leq t \leq t_M(g(s))\}$ and writing $H = hR$ factors out R^2 . $\square$

Proposition 7.2 (Projected area and covering). *For every integer $N \geq 2$, $R, H > 0$ and every $g \in \mathcal{G}$, the top-view projections of the N leaflets cover the disk of radius R with multiplicity one almost everywhere; in particular their areas add up to πR^2 and the unloaded design has zero geometric orifice area.*

Proof. Orient piece 1 by (t, z) and piece 2 oppositely; the projected Jacobians are then $R^2 a(1 + \cos t)/2 \geq 0$ and $R^2 b(1 + \cos t)/2 \geq 0$. For a test function f on the plane, write $f dx \wedge dy = d\alpha$ and apply Stokes to each piece. Summing over pieces and leaflets, the junctions cancel (same curve, opposite orientations), the free edges cancel between neighbouring leaflets (Theorem 5.3), the posts and the base of piece 2 project to points, and the bases of the pieces 1 make up the circle of radius R once, counter-clockwise. Hence $\sum \int f(F) |J| = \int_{\text{disk}} f$, so by the area formula the multiplicity of the projection is 1 almost everywhere on the disk and 0 outside; since the union of the images is compact, no point of the open disk is missed. The profile g only reparametrises the heights and does not change the argument. $\square$

7.1 Keeping R and H fixed

With the paper's linear profile the total area grows with N (Proposition 6.1), so a fair comparison must change something. If R and H are both kept, only the profile $g \in \mathcal{G}$ is free, and the geometry limits what it can do. We take $R = 1$.

Proposition 7.3 (Two leaflets). *For $N = 2$ and every $H > 0$, the linear profile has the least total area in $\mathcal{G}$, uniquely, and every value in $[A_2^{\text{tot}}(H, s), \pi(2H + 1))$ is the total area of some $g \in \mathcal{G}$.*

Proof. For $\theta = \pi$, $t_M \equiv 2\pi$ (Theorem 5.2), so the total area is $\int_{\pi}^{2\pi} \int_0^1 \phi_t(g'(s)) ds dt$ with $\phi_t(p) = \sqrt{4H^2 + p^2(1 + \cos t)^2}$, strictly convex in p for $t \neq \pi$. As $\int_0^1 g' = 1$, Jensen's inequality gives $\int_0^1 \phi_t(g') \geq \phi_t(1)$, with equality only for $g' \equiv 1$. Also $\phi_t(p) \leq 2H + p(1 + \cos t)$, with strict inequality when $p > 0$ and $t > \pi$; since g' is continuous with $\int_0^1 g' = 1$, it is positive on a set of positive measure, so the total area is strictly below $\pi(2H + 1)$. Profiles whose derivative concentrates near one height approach this value. Finally $\mathcal{G}$ is convex and the area is continuous on it (in the C^1 topology), so along segments of $\mathcal{G}$ every intermediate value is attained. $\square$

Reaching the area of the linear $N = 3$ design with $N = 2$ at $H = R$ takes very steep profiles in the families we tried (s^m with $m \approx 17$).

Proposition 7.4 (A lower bound over all profiles). *For $N \geq 3$, $\theta = 2\pi/N$, $c(t) = 1 + \cos t$ and $\mu(t) = |\{z \in [0, 1] : t_M^\theta(z) \geq t\}|$, every $g \in \mathcal{G}$ satisfies*

$$A_N^{\text{tot}}(H, g) \geq LB_N(H) = \frac{N}{2} \left[\int_{\pi}^{\pi+\theta} \sqrt{4H^2 + c^2} dt + \int_{\pi+\theta}^{2\pi} c(t) \mu(t) dt \right].$$

At $H = 1$, LB_N exceeds the total area 8.6809 of the linear $N = 3$ design for $N = 5, 6, 7$ (quadrature, error below 10^{-13}), so no profile gives those N that area.

N	fixed $R = 10$ mm			fixed $H = 15$ mm		
	H (mm)	h	fold angle ($^\circ$)	R (mm)	h	fold angle ($^\circ$)
2	18.63	1.863	0	11.81	1.270	0
3	15.00	1.500	49.2–53.1	10.00	1.500	49.2–53.1
4	12.85	1.285	67.8–82.4	8.77	1.711	75.3–85.5
5	11.43	1.143	75.0–101.5	7.90	1.898	91.4–105.5
6	10.43	1.043	77.4–114.7	7.27	2.063	102.4–118.6
7	9.68	0.968	77.6–124.4	6.79	2.209	110.3–127.7

Table 1: Equal-area family (total area 1230.58 mm^2 , that of the $N = 3$ design with $R = 10$ mm, $H = 15$ mm, the dimensions of the paper’s models). Fold angle: minimum over the junction – supremum (limit at its base).

Proof. For any measurable $\varphi(t) \in [0, \pi/2]$ and $p, c \geq 0$, $\sqrt{4H^2 + p^2c^2} \geq 2H \cos \varphi + pc \sin \varphi$ (Cauchy–Schwarz). Take $\tan \varphi = c/(2H)$ on $[\pi, \pi + \theta]$ and $\varphi = \pi/2$ on $(\pi + \theta, 2\pi]$. Since $t_M(g(s)) \geq \pi + \theta$ for every s , the $2H \cos \varphi$ part integrates over $[\pi, \pi + \theta]$ only. For the other part, the substitution $z = g(s)$ and Fubini give $\int_0^1 g'(s) \int_{\pi}^{t_M(g(s))} c \sin \varphi dt ds = \int_{\pi}^{2\pi} c(t) \sin \varphi(t) \mu(t) dt$. On $[\pi, \pi + \theta]$, $\mu = 1$ and $2H \cos \varphi + c \sin \varphi = \sqrt{4H^2 + c^2}$ at the chosen φ . $\square$

Proposition 7.5 (Four leaflets). *At $H = R$, every $g \in \mathcal{G}$ gives $N = 4$ a total area $> 8.7936 R^2$, while the linear $N = 3$ design has $< 8.6986 R^2$; so no profile gives $N = 4$ the area of the linear $N = 3$ design. At $H = 1.5R$ (the proportions of the paper’s models) an explicit profile does.*

Method. The first statement is a finite certificate in interval arithmetic (30 digits): left and right rectangle sums in z and t , which bound the integrals using the monotonicity of t_M and of $1 + \cos t$; supporting lines of the convex integrand in g' ; and weak duality. It uses no quadrature error estimates and no optimiser. The second is a witness profile evaluated by adaptive quadrature.

So keeping both dimensions fixed forces very different vertical profiles for different N , or is impossible; this supports fixing only one of them.

7.2 The equal-area families

Table 1 gives, for the reference design $N = 3$, $R = 10$ mm, $H = 15$ mm (the dimensions of the paper’s models) and the linear profile, the height (at fixed R) or the radius (at fixed H) that keeps the total area, together with the fold angle along the junction. The *largest* fold angle (at the base of the junction) grows with N in both families for $N = 2, \dots, 7$, more at fixed height; the angle does not grow at every point of the junction. For a flow study this means that changing N at equal area also changes how sharply the surface normal jumps across the junction. The accompanying package provides these surfaces with exact normals, the crease location and meshes.

8 An exploratory mechanical check: sharp versus rounded junction

Does the crease matter mechanically? We solved a deliberately simple problem, not a reproduction of the paper’s LS-Dyna study: one leaflet ($N = 3$, $R = 10$ mm, $H = 15$ mm) as a thin 3D solid (thickness 0.3 mm, quadratic tetrahedra, two layers), St Venant–Kirchhoff with $E = 40$ MPa and $\nu = 0.49$, base and commissure posts clamped, and a uniform follower pressure on the ventricular face increased quasi-statically; no contact, inertia or stays. The sharp leaflet is compared with the G^1 rounding of Section 4. Near the base of the commissure post the rounded section turns by almost 180° within half a millimetre and no shell of this thickness fits, so the same small corner ($z < 1/8$) is removed from both geometries and kept only as a fixed patch when the opening area is computed.

The differences between the rounded and the sharp leaflet (+1.03, +0.27, +0.61% on the three meshes; Table 2) are not shown to converge by the tested meshes and are smaller than the effect of the mesh itself (refining from 40×56 to 56×78 changes the opening area by 1.5–1.9%) and of the removed corner (enlarging it from $z < 1/8$ to $z < 1/4$ changes it by 7.4–7.8%; two versus four layers through the thickness:

	24×33	40×56	56×78	40×56 , cut $z < 1/4$
GOA, sharp (cm ²)	1.968	2.025	2.056	2.175
GOA, rounded (cm ²)	1.988	2.030	2.069	2.189
rounded vs sharp	+1.03%	+0.27%	+0.61%	+0.67%

Table 2: Geometric orifice area at 75 mmHg (exact load level), the highest level reached by all runs.

0.25%). At the resolutions tested we therefore see no effect of the rounding that can be distinguished from discretisation effects. This is not evidence that the crease is mechanically irrelevant, for four reasons.

- The removed corner $z < 1/8$ is where the fold angle is largest and the rounding radius smallest, so the check is blind to the region where the crease is sharpest.
- The comparison level of 75 mmHg was imposed by the runs, not chosen on physical grounds: with full Newton steps, two sharp-leaflet runs stopped before 120 mmHg (76.9 and 98.0 mmHg), and 75 mmHg is the highest level that all runs reached. It is below the 80–120 mmHg range of the paper. (A separate march on the same 40×56 mesh, with different load steps, found converged equilibria up to 120 mmHg; we did not analyse the stops further.)
- On that march, the smallest sampled value of $\det F$ falls from 0.82 at 99.8 mmHg to 0.74 at 119 mmHg: local volume changes of up to about 26% in a material meant to be nearly incompressible ($\nu = 0.49$). Such volume changes raise concerns about the suitability of the St Venant–Kirchhoff model, known to behave poorly under large compression, at those loads; we did not assess its constitutive validity, and $\det F$ was not recorded at 75 mmHg.
- The model has no contact, inertia or stays, and its boundary conditions are our own.

A mechanical conclusion would need at least a smaller removed corner (which the rounded geometry does not allow at this thickness), a hyperelastic model suited to large strains (for instance neo-Hookean) and the paper’s load range.

9 Limitations

- All geometric results (Sections 3–7) concern the unloaded geometry; Section 8 treats the loaded leaflet. Sections 3–6 use the vertical scale $h = H/R = 1$ unless stated. The paper’s models are 20 mm in diameter and 15 mm high; that their shape is our family stretched to $h = 1.5$ (a linear vertical coordinate) is our assumption, not confirmed from the paper. Section 7 and the package treat general h . Only the paper’s linear vertical profile is used, except where the profile is varied explicitly.
- Gauss–Bonnet (Theorem 3.6) and the local rigidity lemma are classical arguments written by hand, not formalised. The numerical integrals (areas, sector angles, concentrated curvature) are floating-point quadratures without interval certification, except the impossibility for $N = 4$ at $H = R$, which is certified in interval arithmetic.
- The aortic-valve papers before 2026 (McKee et al., 2021; Rebolledo et al., 2023; Oliveira et al., 2022, 2024) concern the 2021 model and its elliptic generalisation, not the 2026 construction. The kink of the 2021 model was known and was regularised by Rebolledo et al. (2023); we found no statement there about developability, the size of the fold angle, or an obstruction to a developable rounding. The Wheatley mitral valve (Oliveira et al., 2025) is a separate two-leaflet construction; its circular case uses radii a and $1 - a$, so it shares the relation $a + b = 1$ of 2026, and it has, in the plane, the same two-circle structure as our $N = 2$ case (their a playing the role of our b , up to a reparametrisation of the angle). The paper mentions linear increments in z as one possibility but does not give the three-dimensional parametrisation, so whether our $N = 2$ results apply to it is open.

- The N -leaflet family is our own extension; it need not coincide with the family $G_N(\alpha)$ of the talk. The generalisation of Oliveira et al. (2022) (ellipses, curved walls, opening) does not vary the number of leaflets.
- The mechanical check of Section 8 is exploratory: quasi-static, without contact, inertia or stays, with our own boundary conditions, a St Venant–Kirchhoff material whose suitability at the higher loads is doubtful and was not assessed, and the corner where the crease is sharpest removed. It supports no mechanical conclusion and does not validate or contradict the paper’s results.

Data availability

Code, Lean proofs and the claim table:

<https://github.com/koobzaar/wheatley-two-circle-geometry>. Lean 4 with mathlib (versions pinned in lean/lean-toolchain and lean/lake-manifest.json; `cd lean && lake build`; no sorry, only the standard axioms). Numerics: package `sim/src/wheatley`, tests `uv run --project sim pytest -q`; figures `sim/scripts/technote_figures.py`; symbolic checks of the written proofs `sim/scripts/technote_proof_checks.py`; Section 7: `sim/scripts/a6_*.py`; Section 8: `sim/fem/(FEniCSx)`; the G_N package: `sim/src/wheatley/gn.py` and `deliverable/gn/`. Status and evidence of every result: `CLAIMS.md` and Appendix A.

References

- M. P. do Carmo. *Differential Geometry of Curves and Surfaces*. Prentice-Hall, 1976.
- D. Fuchs and S. Tabachnikov. More on paperfolding. *The American Mathematical Monthly*, 106(1):27–35, 1999. Preprint MPI 95-115.
- P. Hartman and L. Nirenberg. On spherical image maps whose Jacobians do not change sign. *American Journal of Mathematics*, 81:901–920, 1959.
- S. McKee, J. A. Cuminato, I. W. Stewart, and D. J. Wheatley. A mathematical representation of the Wheatley heart valve. *Journal of Biomechanical Engineering*, 143(8):081006, 2021. doi: 10.1115/1.4050753.
- S. McKee, H. L. Oliveira, J. A. Cuminato, E. Seed, C. McCormick, M. Kerr, I. W. Stewart, and D. J. Wheatley. Two circle construction of the reinforced Wheatley valve. *Biomedical Materials & Devices*, 2026. doi: 10.1007/s44174-026-00691-4.
- H. L. Oliveira, S. McKee, G. C. Buscaglia, J. A. Cuminato, I. W. Stewart, and D. J. Wheatley. A generalized mathematical representation of the shape of the Wheatley heart valve and the associated static stress fields upon opening and closing. *IMA Journal of Applied Mathematics*, 87(4):537–567, 2022. doi: 10.1093/imamat/hxac016.
- H. L. Oliveira, G. C. Buscaglia, R. R. Paz, F. Del Pin, J. A. Cuminato, M. Kerr, S. McKee, I. W. Stewart, and D. J. Wheatley. Three-dimensional fluid-structure interaction simulation of the Wheatley aortic valve. *International Journal for Numerical Methods in Biomedical Engineering*, 40(2):e3792, 2024. doi: 10.1002/cnm.3792.
- H. L. Oliveira et al. Mathematical representation and nonlinear modelling of the Wheatley mitral valve. *Medical Engineering & Physics*, 136:104283, 2025. doi: 10.1016/j.medengphy.2025.104283.
- R. J. Rebolledo, S. McKee, J. A. Cuminato, I. W. Stewart, and D. J. Wheatley. Regularization of a mathematical model of the Wheatley heart valve. *Journal of Biomechanical Engineering*, 145(1):014501, 2023.

A Verification table

Each result of the note, the kind of evidence behind it, the Lean theorems (in `lean/WheatleyGeom/`) and the scripts or tests (in `sim/`) that support it. *Lean*: proved in Lean; *numeric*: computed by a versioned script, with a test where listed; *hand*: written argument from classical results, or reading of a source.

ID	Result	Evidence	Lean theorems	Scripts and tests
C-001	Theorem 3.1	Lean	<code>surf1_cone</code> , <code>surf2_cone</code> , <code>surf2_top_circle_unit</code>	<code>test_surface{1,2}_is_cone_with_apex</code>
C-002	Lemma 3.2	Lean	<code>gaussK_cone</code> , <code>gaussK_surf1</code> , <code>gaussK_surf2</code>	<code>test_developable</code>
C-003	Theorem 4.2	Lean	<code>foldCos_junction_three</code> , <code>foldCos_junction_three_bounds</code> , <code>foldCos_eq</code>	<code>test_fold_three_closed_form_and_bounds</code>
C-004	Section 2.2	Lean	<code>surf2_at_zero</code> , <code>surf2_eq_rotZ_smallCircle</code> , <code>map34_E</code> , <code>rotZ_E</code> , <code>map34_eq_rotZ_neg</code>	<code>test_A1_*</code> , <code>test_A2_*</code>
C-005	Section 2.1	Lean	<code>junction_on_circle1</code> , <code>junction_on_circle2</code> , <code>cos_tM</code> , <code>sin_tM</code> , <code>surf1_tM</code> , <code>surf2_tM</code>	<code>test_tM_arcsin_and_arccos_forms_agree_and_hit_M</code>
C-006	Lemma 3.2	Lean	<code>isRegularAt_cone_iff</code> , <code>surf1_isRegularAt_iff</code> , <code>surf2_isRegularAt_iff</code>	–
C-007	Section 2.2	numeric	–	<code>sim/scripts/audit_eqs_27_29.py</code> , <code>test_A4_eq28_needs_sqrt3_factor</code>
C-008	Section 2.2	hand	–	reading of pp. 9–10 of the paper; the statement itself is C-001
C-009	Proposition 3.5	Lean	<code>congruence</code> , <code>congT_isometry</code> , <code>congT_apex1</code> , <code>stay1_length</code> , <code>stay2_length</code> , <code>congruenceθ</code> , <code>congTθ_isometry</code>	<code>test_congruence</code>
C-010	Theorem 3.4	Lean	<code>firstForm_cone_eq_develop</code> , <code>develop_injOn</code> , <code>devAngle_strictMono</code> , <code>devAngle_le</code> , <code>isRegularAt_iff_of_firstForm_eq</code> , <code>firstForm_surf{1,2}_develop</code> , <code>develop{1,2}_injOn</code> , <code>develop{1,2}_isRegularAt_iff</code> , <code>devAngle{1,2}_mem</code>	<code>sim/tests/test_develop.py</code>
C-011	Theorem 3.6	numeric + hand	–	<code>sim/scripts/junction_geodesic_curvature.py</code> , <code>test_junction_geodesic_curvatures_do_not_cancel</code> , <code>test_junction_images_same_length</code>
C-012	Theorem 5.1	Lean	<code>sum_eq_one_of_same_param</code> , <code>$\Delta\theta$_pos</code> , <code>cMθ_sq_add_sMθ_sq</code> , <code>junctionθ</code> , <code>surf1_tMθ</code> , <code>tMθ_two_pi_div_three</code> , <code>tMθ_zero</code> , <code>tMθ_half</code>	<code>sim/tests/test_nleaflet.py</code>

ID	Result	Evidence	Lean theorems	Scripts and tests
C-013	Theorem 5.3	Lean	top_edge_coincide, tM θ _half	test_top_edge_closes
C-014	Theorem 5.2	Lean	foldCos_eq, foldCos_junction, foldCos_junction_lt_one_iff, foldCos_junction_two, dot_cross_w1_w2 θ	test_fold_cos_formula_matches_normals, test_top_fold_angle, test_fold_range_four_leaflets
C-015	Proposition 6.1	numeric; Lean (density)	area_density_sum	nleaflet.leaflet_area, test_area_not_preserved
C-016	Proposition 6.2	Lean + numeric	small_meets_surf1_only_at_E	sim/scripts/nleaflet_separation.py, sim/scripts/coaptation_pairs.py
C-017	Theorem 4.4	Lean + hand	c1_eq, c2 θ _eq, surf1_in_tangent_plane, surf2 θ _in_tangent_plane, no_common_tangent_plane, common_tangent_plane_two	—
C-018	Lemma 4.5	hand + Lean ($\Pi \neq 0$)	secondFormL_cone, L_surf1_ne_zero, L_surf2 θ _ne_zero	—
C-019	Section 4	numeric	fillet, section, fillet_strip_gaussian_curvature	sim/src/wheatley/leaflet_mesh.py, sim/tests/test_leaflet_mesh.py
C-020	Section 8	numeric	—	sim/scripts/make_leaflet_meshes.py, sim/fem/leaflet_pressure.py, sim/fem/leaflet_limit.py, sim/scripts/fem_summary.py, sim/scripts/fem_compare.py, sim/out/fem/summary.csv, sim/out/fem/compare.txt
C-021	Proposition 4.3	Lean	foldCosH_eq, foldCosH_junction_three, foldCosH_three_bounds_h32	sim/tests/test_vertical_scale.py
C-022	Section 1	hand	—	reading of the cited papers (quotations are verbatim)
C-024	Proposition 7.1	numeric	—	test_density_formula
C-025	Proposition 7.3	hand	—	test_two_leaflets_linear_profile_minimises_area
C-027	Proposition 7.2	hand	—	test_swept_area_is_pi_over_N, test_projected_jacobians_sum
C-028	Proposition 7.3	hand	—	—
C-029	Proposition 7.4	hand + numeric	—	sim/scripts/a6_lower_bound.py
C-030	Proposition 7.5	numeric (certified)	—	sim/scripts/a6_certificate_n4.py, sim/out/a6/certificate_n4.txt, sim/scripts/a6_min_area.py, sim/out/a6/witness_*.json

ID	Result	Evidence	Lean theorems	Scripts and tests
C-031	Proposition 7.1	numeric	–	test_area_scaling_and_reference
C-032	Section 7.2	numeric	–	sim/scripts/a6_equal_area_family.py, sim/out/a6/equal_area_family.txt, deliverable/gn/table_fixed_{R,H}.csv, test_equal_area_families
C-033	Section 7.2	numeric	–	sim/src/wheatley/gn.py, sim/tests/test_gn.py, sim/scripts/gn_export.py